

Experiment 2: Fixed Height + Fixed Wind + Obstacle Ablation | random\_obstacle | Comparison Table

Dropped-Only Quality

| Method  | N(drop) | Precision Mean(m) | Precision P50(m) | Precision P90(m) | R@1m  | R@2m  | R@3m  | Roll  Mean | Roll  P90 | Pitch  Mean | Pitch  P90 | Impact Mean | Impact P90 | Impact P95 |
|---------|---------|-------------------|------------------|------------------|-------|-------|-------|------------|-----------|-------------|------------|-------------|------------|------------|
| PPO     | 1994    | 2.596             | 1.418            | 6.145            | 33.1% | 67.3% | 79.6% | 14.629     | 31.234    | 9.432       | 24.818     | 0.583       | 1.333      | 1.477      |
| PPO-GRU | 1991    | 3.020             | 0.745            | 13.403           | 63.9% | 81.2% | 83.4% | 6.907      | 16.432    | 5.975       | 13.143     | 0.343       | 0.735      | 1.002      |
| PPO-RNN | 1983    | 2.041             | 1.082            | 3.512            | 46.4% | 78.7% | 88.6% | 12.459     | 24.427    | 6.097       | 12.458     | 0.413       | 1.056      | 1.386      |
| MPC     | 1063    | 5.817             | 3.070            | 15.549           | 3.8%  | 19.0% | 47.2% | 2.851      | 3.676     | 1.640       | 2.841      | 0.163       | 0.147      | 0.210      |

No-Drop Breakdown (done but no release)

| Method  | N(done) | N(drop) | Drop Rate | N(no_drop) | No-drop Rate | Crash Rate | Crash+NoDrop | Crash in NoDrop | Timeout+NoDrop | Both Flags |
|---------|---------|---------|-----------|------------|--------------|------------|--------------|-----------------|----------------|------------|
| PPO     | 2000    | 1994    | 99.7%     | 6          | 0.3%         | 0.3%       | 6            | 100.0%          | 0              | 0          |
| PPO-GRU | 2000    | 1991    | 99.5%     | 9          | 0.5%         | 0.5%       | 9            | 100.0%          | 0              | 0          |
| PPO-RNN | 2000    | 1983    | 99.2%     | 17         | 0.8%         | 0.8%       | 17           | 100.0%          | 0              | 0          |
| MPC     | 2000    | 1063    | 53.1%     | 937        | 46.9%        | 46.9%      | 937          | 100.0%          | 0              | 0          |

Cross-Case Check: PPO/PPO-GRU/PPO-RNN on MPC No-Drop Cases (large error > 3.0m)

| Method  | In MPC-NoDrop | Dropped | Drop Rate | Err>3.0m | LargeErr Rate (in dropped) | Mean Err (in dropped) |
|---------|---------------|---------|-----------|----------|----------------------------|-----------------------|
| PPO     | 937           | 934     | 99.7%     | 239      | 25.6%                      | 2.868                 |
| PPO-GRU | 937           | 928     | 99.0%     | 144      | 15.5%                      | 2.959                 |
| PPO-RNN | 937           | 927     | 98.9%     | 93       | 10.0%                      | 1.961                 |